

SVI Calibration of the SPX Implied Volatility Smile: Sequential Fitting, Ridge Penalisation and Calibration Diagnostics

Abstract

We calibrate the SVI model $w(k) = a + b \left[\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right]$ to 103 daily snapshots of S&P 500 index option implied volatilities for a single expiry (18 December 2026), targeting mid implied volatilities in total-variance space. Fits run sequentially in seven parallel tracks spanning $\lambda \in \{0, 10^{-7}, \dots, 10^{-2}\}$, where λ weights a scale-normalised ridge penalty on day-on-day parameter changes, with identical multistart budgets so that tracks are directly comparable. The penalty weight is located empirically: fit error reaches its minimum at $\lambda = 10^{-5}$, where parameter-path roughness falls by roughly three quarters relative to independent fitting while mean fit error *improves* (27.0 vs 29.0 vol bps). Two data-quality findings emerge: the exported call and put implied volatilities are not parity-consistent, so the forward must be recovered as a minimiser rather than a parity root; and the bid-ask violation penalty of the specification cannot be implemented, as the export contains no bid or ask quotes.

1 Data and cleaning

Underlying / source	S&P 500 index options; Bloomberg Terminal
Snapshots	103 daily snapshots, January - June 2026
Expiry / time to expiry	18 December 2026 (single expiry); T from 0.96 down to 0.55 years
Fields supplied	date, expiry, strike, call/put flag, implied volatility
Fields absent	bid/ask prices and IVs, spot, forward, discount curve
Rows	44,234 raw $\rightarrow$ 42,160 after cleaning
Filters	drop missing IVs (2,074 rows); $T > 0$; OTM side only; per-day adaptive moneyness window (central 85 % of quotes by $ k $)
Quotes fitted per day	182 on average

Table 1: Data provenance and cleaning filters.

The raw export is moderately clean but not usable as delivered. Missing implied volatilities (4.7% of rows) concentrate in the deep wings where no usable quote exists, and the surviving quotes immediately outside that region exceed 100% implied volatility. The strike grid is regular and calls and puts are matched at every strike. With no bid/ask fields, the zero-bid and stale-price filters the specification anticipates cannot be applied directly; the adaptive moneyness window serves as the liquidity filter instead, since illiquidity manifests here as missing and erratic quotes rather than as observable spreads. The single implied-volatility column is taken to be the Bloomberg mid implied volatility, which the specification permits as a direct calibration target; no mid can be reconstructed independently.

2 Method

Quotes are mapped to log-moneyness $k = \ln(\frac{K}{F})$ and total implied variance $w = IV^2 T$, retaining out-of-the-money contracts only (puts for $k \leq 0$, calls for $k > 0$). For each snapshot,

$$\mathcal{L}(\theta_t) = \underbrace{\frac{1}{n} \sum_i \frac{(w_{\text{SVI}}(k_i; \theta_t) - w_i)^2}{1 + 5k_i^2}}_{\text{fit to mid}} + \underbrace{10^3 \min(\min_k w_{\text{SVI}}(k; \theta_t), 0)^2}_{\text{non-negative variance}} + \underbrace{\lambda \sum_j \left(\frac{\theta_{t,j} - \theta_{t-1,j}}{s_j} \right)^2}_{\text{ridge on parameter changes}} \quad (1)$$

with $s = (0.05, 0.5, 0.5, 0.2, 0.2)$; the scaling is essential because the parameters differ by more than an order of magnitude, and without it λ would regularise only the numerically largest. The vega-mimicking weight concentrates the fit near the money, where premia are most sensitive to volatility. Optimisation is multistart L-BFGS-B (8 starts, fixed seed, fully reproducible) with box bounds $b > 0$, $|\rho| < 1$, $\sigma > 0$, warm-started from the previous snapshot.

Forward estimation and a parity anomaly. Put-call parity implies $IV_c(F) = IV_p(F)$, so the forward should be the zero of $d(K) = IV_c(K) - IV_p(K)$. In this dataset $d(K)$ has *no zero crossing*: it is negative at every strike (-0.0063 to -0.1379 on the first snapshot), indicating the exported volatilities embed a discounting or dividend assumption that does not reconcile the two legs. The forward is therefore taken as the minimiser of $|d(K)|$, estimated at sub-grid resolution by fitting a parabola to the smoothed curve near its minimum, subject to a continuity constraint against the previous day. This matters materially: a naive grid-based minimiser returns forwards quantised to listed strikes, injecting one grid step ($\approx 1.4\%$) of spurious daily variation and manufacturing apparent index moves of up to 16%. After correction, the median absolute daily change falls to 0.68% and 7 estimates remain flagged as discontinuous.

3 Fit quality across the sample

Figure 1 shows the two ends of the fitting problem. The first snapshot (02 January 2026) has no history - no warm start, no ridge anchor - so the fit rests entirely on the multistart search, which includes a skew-informed guess ($\rho_0 = -0.6$); the SPX smirk is reproduced immediately (RMSE 24.7 bps) and every later snapshot inherits this solution as its starting point. The best-fitting snapshot (05 January 2026, 23.0 bps), selected among non-event, non-flagged dates so that the clean-smile illustration cannot coincide with a disturbance, shows five parameters reproducing roughly 182 quotes to within a few basis points through the body of the smile; mid-sample fits are visually identical between penalised and unpenalised tracks.

The fitted surface captures the SPX skew throughout: in the ridge track ρ averages -0.379 and never approaches zero, holding the steep left wing that prices out-of-the-money crash protection (the unpenalised track's excursions to positive ρ in late May are the instability discussed in Section 4, not a feature of the market). Residuals are, as the specification anticipates, larger for deep out-of-the-money strikes, though the effect is modest: pooled across snapshots, RMSE is 27.8 bps within $|k| \leq 0.15$ against 32.1 bps for $|k| > 0.35$, an increase of roughly 15%. Vega vanishes in the wings, so a volatility discrepancy there is a negligible premium error, and forcing the wings to fit would distort the liquid core. The pattern is also structural, since SVI's asymptotically linear wings cannot match the market's mild wing convexity. Both far wings sit slightly below the market in every fit for the same reason.

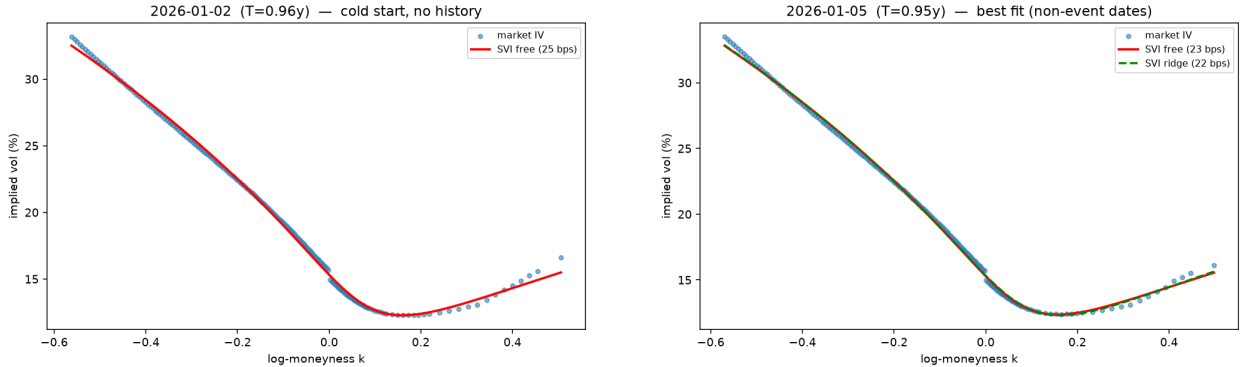


Figure 1: Cold-start fit on the first snapshot (left) and best-fitting non-event snapshot (right).

4 Parameter dynamics and the choice of λ

The unpenalised track answers the specification's first stability question directly: it jumps far more between snapshots than the fitted curves do. The same smile is represented by materially different parameter vectors on consecutive days - weak identification, since shifting m while compensating through a and b leaves the curve nearly unchanged - compounded by occasional excursions into inferior local minima, visible as isolated spikes in Figure 2. This movement is an artefact of the calibration, not information about the market, and the instability originates where quotes are weakest: the deep wings, where the wing parameters b and σ are least constrained. The adaptive moneyness window is designed to attack this at source, excluding the outer 15% of quotes by $|k|$ so that the least reliable observations never enter the objective; data filtering and regularisation are therefore complementary rather than substitutes.

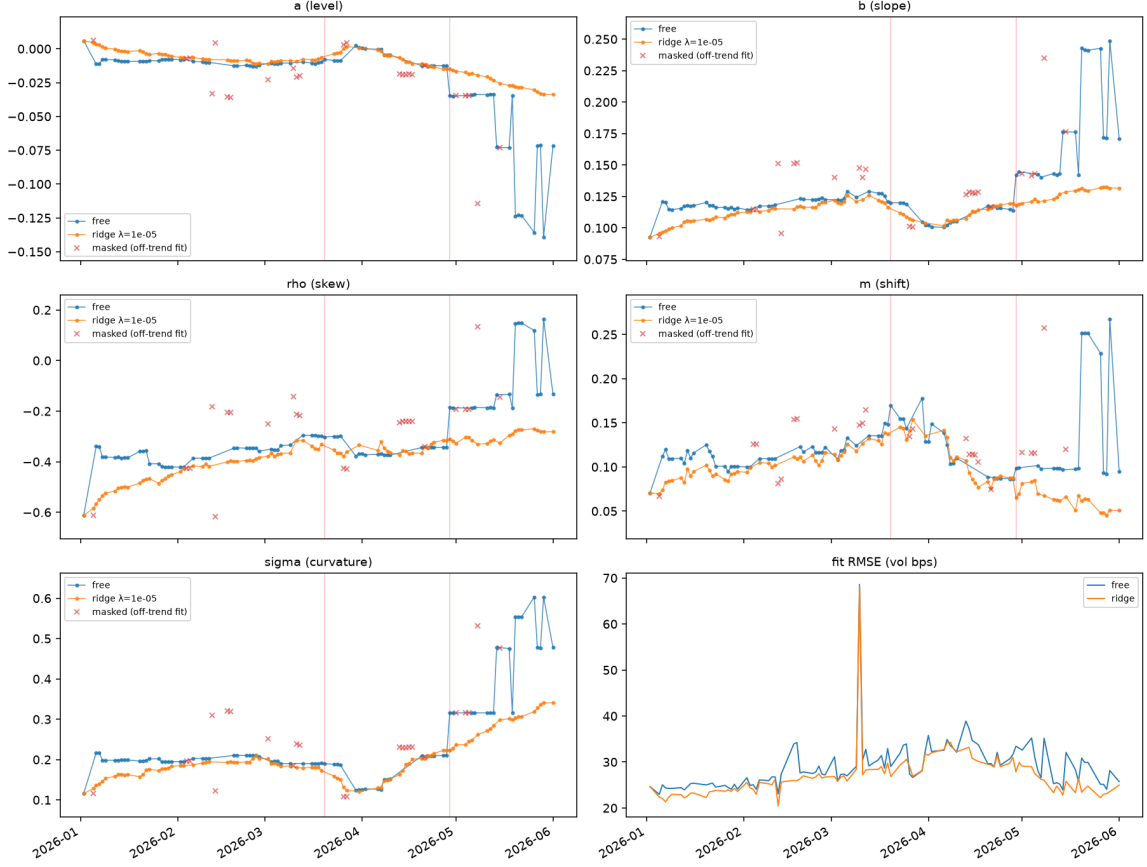


Figure 2: Full-sample parameter paths, unpenalised versus ridge ($\lambda = 10^{-5}$), with per-day fit RMSE. Vertical lines mark event dates. Isolated off-trend fits (24 free, 12 ridge; marked $\times$) are excluded from the connecting lines so the paths read continuously; the criterion is six times the rolling MAD, floored at 5 % of each parameter’s standard deviation. All statistics use the complete series.

Ridge penalisation smooths what remains, substantially: day-on-day change volatility falls from 0.1092 to 0.0187 for the skew parameter and from 0.0625 to 0.0069 for curvature (Table 2), with aggregate scaled roughness down from 0.274 to 0.068 (Table 3).

Parameter	Unpenalised ($\lambda = 0$)				Ridge ($\lambda = 10^{-5}$)			
	mean	min	max	std(Δ)	mean	min	max	std(Δ)
a (level)	-0.0231	-0.1393	0.0060	0.0207	-0.0099	-0.0339	0.0060	0.0010
b (slope)	0.1319	0.0926	0.2487	0.0242	0.1154	0.0926	0.1326	0.0024
ρ (skew)	-0.2923	-0.6167	0.1648	0.1092	-0.3792	-0.6123	-0.2569	0.0187
m (shift)	0.1233	0.0668	0.2677	0.0419	0.0976	0.0398	0.1784	0.0139
σ (curvature)	0.2459	0.1080	0.6031	0.0625	0.1984	0.1155	0.3406	0.0069

Table 2: Fitted SVI parameters across 103 snapshots; std(Δ) is the day-on-day change volatility, the stability measure.

How large can the penalty be before the fit suffers? The sweep locates the boundary precisely (Table 3): mean fit error reaches its minimum at $\lambda = 10^{-5}$ and rises monotonically thereafter - by 5% at 10^{-4} , 42% at 10^{-3} and 100% at 10^{-2} - while roughness falls monotonically across the entire range. Degradation therefore begins immediately above 10^{-5} and is severe by 10^{-3} . One detail is worth recording: the weakest penalty tested, 10^{-7} , fits marginally *worse* than the baseline (29.2 vs 29.0 bps) despite identical search budgets - a penalty too weak to guide the optimiser can still perturb it slightly away from solutions the free search would reach.

λ	mean RMSE (vol bps)	mean scaled $\ \Delta\theta\ $	regime
0	29.0	0.2743	baseline (unpenalised)
10^{-7}	29.2	0.1303	penalty inert
10^{-6}	27.6	0.1054	mild smoothing
10^{-5}	27.0	0.0675	selected: optimum on both axes
10^{-4}	28.5	0.0391	smoother, small fit cost
10^{-3}	38.4	0.0163	over-smoothed
10^{-2}	54.0	0.0078	fit degraded

Table 3: Stability-accuracy trade-off; all tracks share an identical multistart budget.

Does ridge penalisation improve stability without over-smoothing? At $\lambda = 10^{-5}$, yes, and the evidence is unusually clean: the selected weight improves *both* axes, so no accuracy is traded for stability at all. Two mechanisms combine - the penalty suppresses parameter movement the data cannot distinguish, and anchoring each optimisation near the previous solution steers the solver away from inferior local minima. That fit quality improves rather than degrades is direct evidence that the suppressed movement carried no information; rising error beyond 10^{-5} marks the point where genuine signal begins to be suppressed too, making the error minimum a principled stopping point rather than a matter of taste. A practitioner prioritising stability could defend 10^{-4} (a further 42% roughness reduction for 5% more error); 10^{-3} and above are unambiguously over-smoothed.

Finally, on whether the fitted surface is smooth across strikes, expiries and time: across strikes, yes - SVI is analytic in k by construction, and Figures 1 shows no kinks or oscillations within the fitting window. Across time, yes at $\lambda = 10^{-5}$, which is what Figure 2 demonstrates and what the roughness column of Table 3 quantifies. Across expiries the question cannot be posed here, since the export contains only one.

5 Market regimes and events

The sample's largest apparent move - close to -9% in early February, visible in Figure 3 - carries no event marking because that date's forward estimate was itself flagged as discontinuous, and flagged dates are excluded from candidacy.

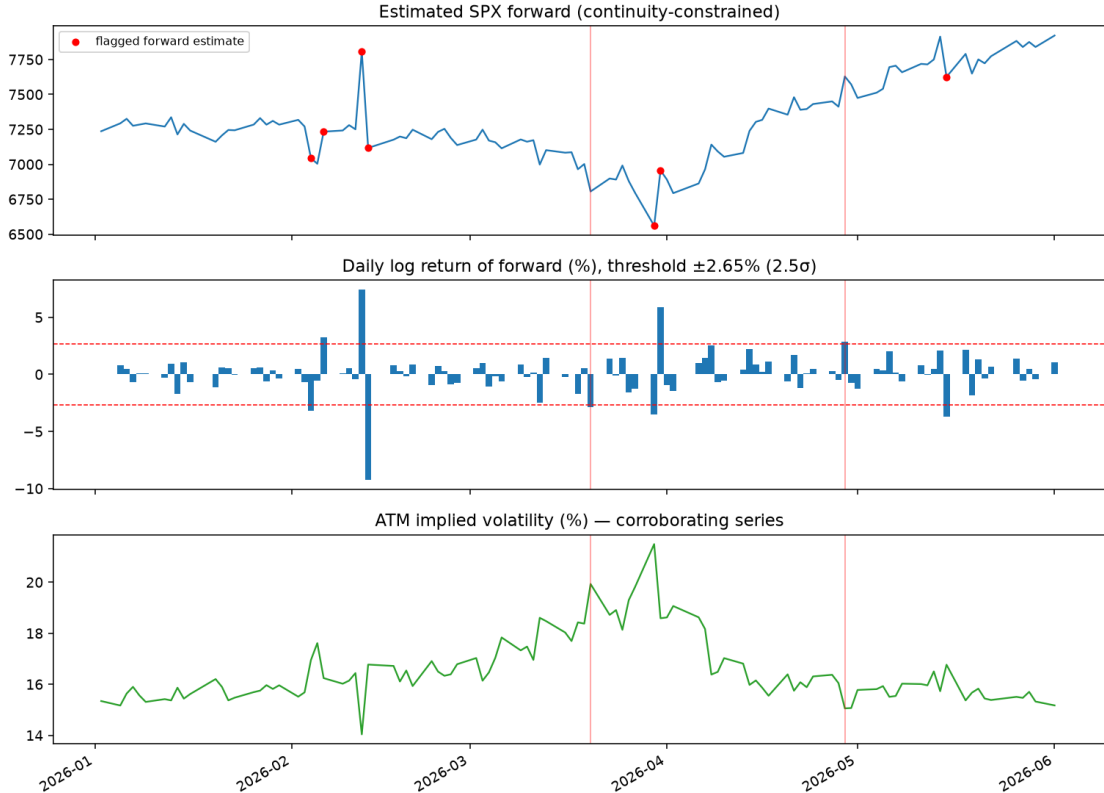


Figure 3: Estimated forward with flagged estimates (top), daily forward log returns against the 2.5σ threshold (middle), ATM implied volatility as the corroborating series (bottom). Vertical lines mark event dates.

The export contains no index level, so daily returns are proxied by $r_t = \ln(\frac{F_t}{F_{t-1}})$ from the estimated forward, and an event is recognised only when a large return is corroborated by a large change in ATM implied volatility - a forward move with no volatility repricing is more likely an estimation artefact than a genuine index move. The proxy's realised standard deviation is 1.80% (threshold 2.65% at 2.5σ), which remains above plausible S&P 500 daily volatility: residual estimation noise persists even after the sub-grid correction, a direct consequence of recovering the forward from a shallow minimum rather than a parity root. Identified event dates - 29 April 2026, 20 March 2026 (basis: return and ATM-vol corroborated) - should therefore be read as candidates derived from option data alone; validation against an external SPX price series is the natural next step. It is noted that 20 March 2026 is a quarterly expiry, on which mechanical flows can move option quotes without a comparable index move.

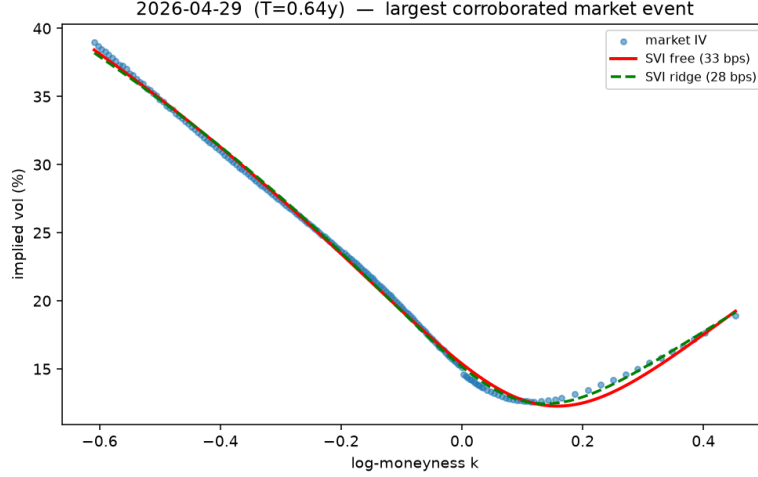


Figure 4: Smile fit of the largest candidate event

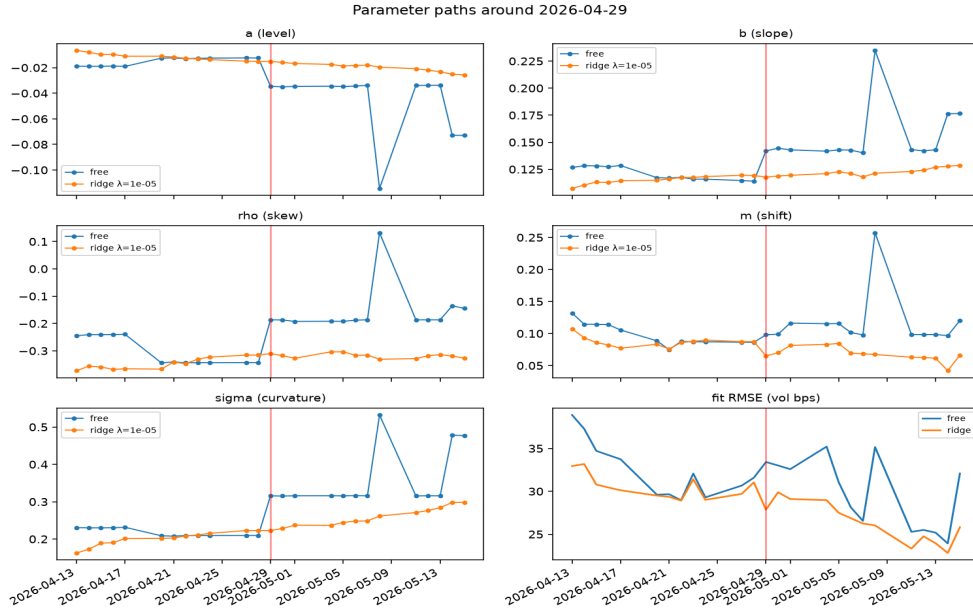


Figure 5: Parameter paths ± 12 trading days around the largest event

The specification asks whether parameters evolve smoothly in calm periods and sharply in volatile ones. The expected pattern is *not* observed - and the reason is instructive. In the ridge track the mean absolute daily change in ρ is 0.0156 outside event windows against 0.0076 within ± 5 days of an event: movement is, if anything, *smaller* near the identified events. Two factors explain the inversion. The sample contains no verified large market shock, so the volatile-regime side of the comparison may simply be absent; and calm-period movement is dominated by residual identification noise, inflating the benchmark against which events are compared. The honest conclusion is that with a single expiry, a five-month window and no verified shock, this question cannot be answered affirmatively from this dataset. What Figure 5 shows is more striking than

mere survival of the event. On the event date itself - where anchoring to the previous day should be most costly - the ridge fit is *better* than the free fit (27.9 against 33.4 vol bps). The free track appears to have mis-converged precisely when the smile was moving, diving to $a \approx -0.10$ and $\rho \approx +0.1$ in the days that follow, while the ridge track transitions smoothly to the new level. Far from preventing repricing, the anchor supplied a good starting point on the day it was hardest to find one. Relatedly, the short-dated regime the specification asks about ($T < 0.1$, where smile curvature is extreme) is never reached here: T runs only from 0.96 to 0.55 years, and within that range fit error shows no systematic deterioration as T shortens.

6 Conclusion

Fitting SVI to each snapshot independently produces parameters that jump around far more than the smiles they describe. A ridge penalty on day-on-day parameter changes fixes this, provided the weight is chosen from the data rather than assumed. Sweeping λ across five orders of magnitude gives a clear answer: at $\lambda = 10^{-5}$, parameter-path roughness falls by roughly three quarters and mean fit error *improves*, from 29.0 to 27.0 vol bps. Nothing is traded away. Above that value the fit deteriorates quickly, doubling in error by 10^{-2} .

Three points follow. First, the ridge penalty is not acting in isolation; the selected moneyness window dictates which quotes influence the objective function, heavily driving the parameters b and σ . Second, rigorous data cleaning must precede model fitting. Discrepancies between exported call and put implied volatilities across strikes caused a grid-based forward estimator to generate phantom index moves of up to 16%. Third, hypotheses comparing calm versus volatile regimes could not be evaluated, as the dataset contains no verified market shocks.

Finally, because the dataset is limited to a single expiration date and lacks bid-ask quotes, this analysis is restricted from testing term-structure dynamics or evaluating bid-ask arbitrage penalties.